

Kian Maleki, PhD — Data Scientist | Machine Learning | Statistical Modeling

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Authorized to work in the United States; no sponsorship required

University of Iowa (2025) PhD in Physics (GPA: 4.03) | Creighton University (2017) MS in Physics (GPA: 4.00)

Profile

- Data scientist with a physics PhD and ten years of research experience in statistical inference, mathematical modeling, and scientific computing.
- Builds end-to-end: open-source Python packages, automated testing and CI, containerized and reproducible pipelines from raw data to evaluated model.
- Author of two single-author machine-learning methodology manuscripts (on classifier evaluation and learning-curve prediction) currently in review at *Machine Learning* and *Pattern Recognition*, alongside peer-reviewed physics publications.

Technical Expertise

- **Machine Learning & Statistics:** Bayesian inference, Monte Carlo simulation, classifier evaluation, calibration, bootstrap confidence intervals, XGBoost, scikit-learn, PyTorch, TensorFlow.
- **Data & Software Engineering:** Python, SQL, Pandas, Polars, NumPy, Numba, FastAPI, Pydantic, Linux/Bash, HPC.
- **MLOps & Cloud:** Docker, Git, GitHub Actions, MLflow, DVC, Evidently, AWS SageMaker, S3, Lambda, CloudWatch, IAM.
- **Generative AI & NLP:** RAG, Transformers, document intelligence, Bedrock.

Training & Certifications

BlueDot Impact | Technical AI Safety Project

Aug 2026 – present

- Developing *Decided Mid-Thought*, a pre-registered study of chain-of-thought faithfulness in reasoning models, as the program research project.

AI-Powered Document Intelligence Extern | Pfizer-Sponsored Externship

Apr 2026 – Jul 2026

link: <https://github.com/drkianmaleki/PharmaDoc-AI>

- Built PharmaDoc-AI, the externship capstone: a prototype automating document extraction, retrieval, and question answering for healthcare supply-chain use cases with native PDF text extraction, LLMs, and RAG architectures.

TripleTen | AI & Machine Learning Program (36 weeks)

Nov 2025 – Apr 2026

- Completed 15+ end-to-end ML projects spanning supervised and unsupervised learning, deep learning, computer vision, NLP, model evaluation, SQL, and Git.

Professional Experience

Independent Machine Learning Engineer | Open-Source Research & Development

Aug 2025 – Present

- Design and ship end-to-end open-source machine-learning software as tested Python packages with pytest, CI via GitHub Actions, and reproducible experiment pipelines: a classifier-evaluation toolkit, a learning-curve benchmarking harness, a RAG document assistant, and a domain-specific product-name normalization engine.
- Completed a pre-registered causal study of chain-of-thought faithfulness in a reasoning model (*Decided Mid-Thought*), using prefix interventions and 500 resampled continuations to test when the final answer becomes determined.
- Engineered a reproducible benchmark evaluating 51 sequence-acceleration methods from 13 families across 18 convergence regimes for finite-horizon learning-curve prediction in gradient boosting, with automated failure detection and conservative extrapolation rules for operational use.
- Implemented ambiguity metrics, calibration analysis, bootstrap confidence intervals, and paired statistical tests to quantify when model comparisons and predictions can be trusted.
- Authored two single-author manuscripts from this work, submitted to *Machine Learning* and under review at *Pattern Recognition*.

Creighton University | Resident Assistant Professor of Physics

Aug 2025 – Jul 2026

- Taught undergraduate physics courses (40–180 students), delivering lectures on optics and electronics with a focus on applied problem-solving.
- Supervised and mentored 1–5 teaching assistants, improving instructional quality and consistency in grading.
- Developed interdisciplinary lab curriculum integrating AI tools and chatbots into scientific workflows.

- Optimized large-scale numerical simulations, reducing computational runtime from days to under one hour through algorithmic improvements.
- Developed high-dimensional modeling frameworks for complex physical systems using statistical and numerical methods.
- Analyzed high-dimensional experimental data to extract signal from noise using statistical inference and optimization.

Selected Projects

Decided Mid-Thought: Locating Answer Commitment in Reasoning Models 2026

Tech: Python, LLM APIs (OpenRouter/DeepInfra), Resampling Experiments, Pre-registered Study Design

link: <https://github.com/drkianmaleki/decided-mid-thought>

- Designed and pre-registered a causal study of whether a reasoning model's visible chain of thought reveals when its final answer becomes determined, using prefix interventions and 500 resampled continuations.
- Found no single-sentence commitment point against pre-registered thresholds; the default answer was ~80% determined by the prompt alone, with 84.6% of the trace generated after the answer had saturated.

LumberLex: Domain-Specific Product Normalization Engine 2026

Tech: Python, RapidFuzz, Pydantic, Streamlit, pytest, Data Normalization, LLM Explanations

link: <https://github.com/drkianmaleki/LumberLex>

- Built a Python package and Streamlit application that normalizes inconsistent lumber product names into canonical structured outputs with confidence scores and manual-review flags, designed to solve a product-data normalization problem described by managers at a building-materials software company.
- Implemented fuzzy matching, dimension extraction, treatment detection, seller identification, and alternative matches, exposed through a Streamlit interface with chatbot-style explanations.

End-to-End MLOps Pipeline: Heart Disease Classification 2026

Tech: DVC, Git, MLflow, pytest, GitHub Actions, Evidently

link: <https://github.com/drkianmaleki/Heart-Disease-Prediction>

- Built a reproducible training-and-deployment workflow with data versioning (DVC), experiment tracking and a model registry (MLflow), and CI/CD validation gates.
- Automated testing with GitHub Actions and pytest, enforcing model-performance thresholds before deployment.
- Implemented data and concept drift monitoring with Evidently to support proactive retraining triggers.

PDF Document Intelligence Extractor 2026

Tech: Python, NLP, Regex, Pandas, PDF Text Extraction, RAG

link: <https://github.com/drkianmaleki/PDF-Document-Intelligence-Extractor>

- Built a document intelligence pipeline to extract key regulatory information from PDF documents using native text-layer parsing rather than OCR.
- Developed rule-based and NLP extraction methods to identify, normalize, and structure information from semi-structured text.

Selected Publications

- Maleki, K. (2026). "Finite-Horizon Learning-Curve Prediction for Gradient Boosting: Why Sequence Acceleration Fails and Simple Curve Fits Suffice." *Machine Learning*, submitted.
- Maleki, K. (2026). "Quantifying Decision-Boundary Uncertainty in Binary Classifiers: The Ambiguity Range Framework." *Pattern Recognition*, under review. Preprint: <https://ssrn.com/abstract=7118401>.
- Maleki, K. et al. "Crystal fields, exchange and dipolar interactions..." *Physical Review B*, 2025.
- Kavand, M., Maleki, K., et al. "A General and Modular Approach to Solid-State Integration..." *Nano Letters*, 2025. Related to a patent-pending quantum device.